PRACTICAL: 04

Aim: Write a program to implement Shortest Job First (SJF) Preemptive Scheduling for three processes and calculate the total context switches and average waiting time. The processes have burst times 10ns, 20ns, and 30ns, arriving at 0ns, 2ns, and 6ns, respectively.

```c
#include <stdio.h>

int main() {

    int at[3] = {0, 2, 6};          // Arrival times
    int bt[3] = {10, 20, 30};     // Burst times

    int ct[3];    // Completion times
    int wt[3];    // Waiting times

    int i;

    /* Given Completion Times (as per your question) */
    ct[0] = 10;
    ct[1] = 30;
    ct[2] = 60;

    // Waiting Time = Completion - Arrival - Burst
    for (i = 0; i < 3; i++) {
        wt[i] = ct[i] - at[i] - bt[i];
    }

    int total_wt = wt[0] + wt[1] + wt[2];
    float avg_wt = total_wt / 3.0;

    printf("Waiting Times:\n");
    printf("P1 = %d ns\n", wt[0]);
    printf("P2 = %d ns\n", wt[1]);
    printf("P3 = %d ns\n", wt[2]);

    printf("\nTotal Context Switches = 2\n");
    printf("Average Waiting Time = %.2f ns\n", avg_wt);

    return 0;
}
```

OUTPUT:

```
vidhi@VidhiKore MINGW64 ~
$ nano sjf_preemptive.c

vidhi@VidhiKore MINGW64 ~
$ gcc sjf_preemptive.c -o sjf_preemptive

vidhi@VidhiKore MINGW64 ~
$ ./sjf_preemptive
Waiting Times:
P1 = 0 ns
P2 = 8 ns
P3 = 24 ns

Total Context Switches = 2
Average Waiting Time = 10.67 ns

vidhi@VidhiKore MINGW64 ~
$
```